Parallel Computing
Strassen Algorithm

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Matrix-Matrix Multiplikation

Input: Two $n \times n$ matrices A and B, where n is an exact power of 2.

Output: The $n \times n$ matrix C given by

$$C \leftarrow A \cdot B.$$

Naiver Algorithmus

- Drei geschachtelte Schleifen
- Laufzeit $T(n) = 2n^3 - n^2$

Block-orientierter Algorithmus

- Partitionierung in vier Block-Matrizen

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

$$A_{ij} \text{ is } n/2 \times n/2$$

- Rekursion

$$C_{ij} = A_{i1} \cdot B_{1j} + A_{i2} \cdot B_{2j}$$

$$\text{for } i, j \in \{1, 2\}$$

Beobachtung

- Acht rekursive Aufrufe pro Level
- Vier $n/2 \times n/2$ Matrix-Matrix Additionen
- Laufzeit:

$$T(n) = 8\,T(n/2) + n^2 \qquad \text{for } n > 1$$

$$T(1) = 1$$

Strassen‘s Algorithm

$$C = A \cdot B$$

$$= \begin{bmatrix} M_1 + M_2 + M_3 - M_4 & M_4 + M_6 \\ M_3 + M_5 & M_1 - M_5 + M_6 + M_7 \end{bmatrix}$$

$$\begin{aligned}
M_1 &= (A_{11} + A_{22}) \cdot (B_{11} + B_{22}) \\
M_2 &= (A_{12} - A_{22}) \cdot (B_{21} + B_{22}) \\
M_3 &= A_{22} \cdot (B_{21} - B_{11}) \\
M_4 &= (A_{11} + A_{12}) \cdot B_{22} \\
M_5 &= (A_{21} + A_{22}) \cdot B_{11} \\
M_6 &= A_{11} \cdot (B_{12} - B_{22}) \\
M_7 &= (A_{21} - A_{11}) \cdot (B_{11} + B_{12}).
\end{aligned}$$

Algorithm 4.13: If $A, B \in \mathbb{R}^{n \times n}$ where n is an exact power of 2, this algorithm computes the matrix-matrix-multiplication $C = A \cdot B$ by applying Strassen's algorithm recursively down to the level where the block size is as small as n_0.

Function $C = \textsc{Strassen}(A, B, n, n_0)$

  if $n > n_0$ **then**

    Partition $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ and $B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$ into blocks of size $n/2 \times n/2$.

    $M_1 \leftarrow \textsc{Strassen}(A_{11} + A_{22}, B_{11} + B_{22}, n/2, n_0)$

    $M_2 \leftarrow \textsc{Strassen}(A_{12} - A_{22}, B_{21} + B_{22}, n/2, n_0)$

    $M_3 \leftarrow \textsc{Strassen}(A_{22}, B_{21} - B_{11}, n/2, n_0)$

    $M_4 \leftarrow \textsc{Strassen}(A_{11} + A_{12}, B_{22}, n/2, n_0)$

    $M_5 \leftarrow \textsc{Strassen}(A_{21} + A_{22}, B_{11}, n/2, n_0)$

    $M_6 \leftarrow \textsc{Strassen}(A_{11}, B_{12} - B_{22}, n/2, n_0)$

    $M_7 \leftarrow \textsc{Strassen}(A_{21} - A_{11}, B_{11} + B_{12}, n/2, n_0)$

    $C \leftarrow \begin{bmatrix} M_1 + M_2 + M_3 - M_4 & M_4 + M_6 \\ M_3 + M_5 & M_1 - M_5 + M_6 + M_7 \end{bmatrix}$

  else

    `// For small matrices use some matrix-matrix-multiplication algorithm`

    $C \leftarrow A \cdot B$

Strassen‘s Beobachtung

- Sieben rekursive Aufrufe pro Level
- Achtzehn n/2 × n/2 Matrix-Matrix Additionen
- Laufzeit

$$T(n) = 7\,T(n/2) + \frac{9}{2}n^2 \qquad \text{for } n > 1$$

$$T(1) = 1$$

$$T(n) = \Theta\left(n^{\log_2 7}\right)$$